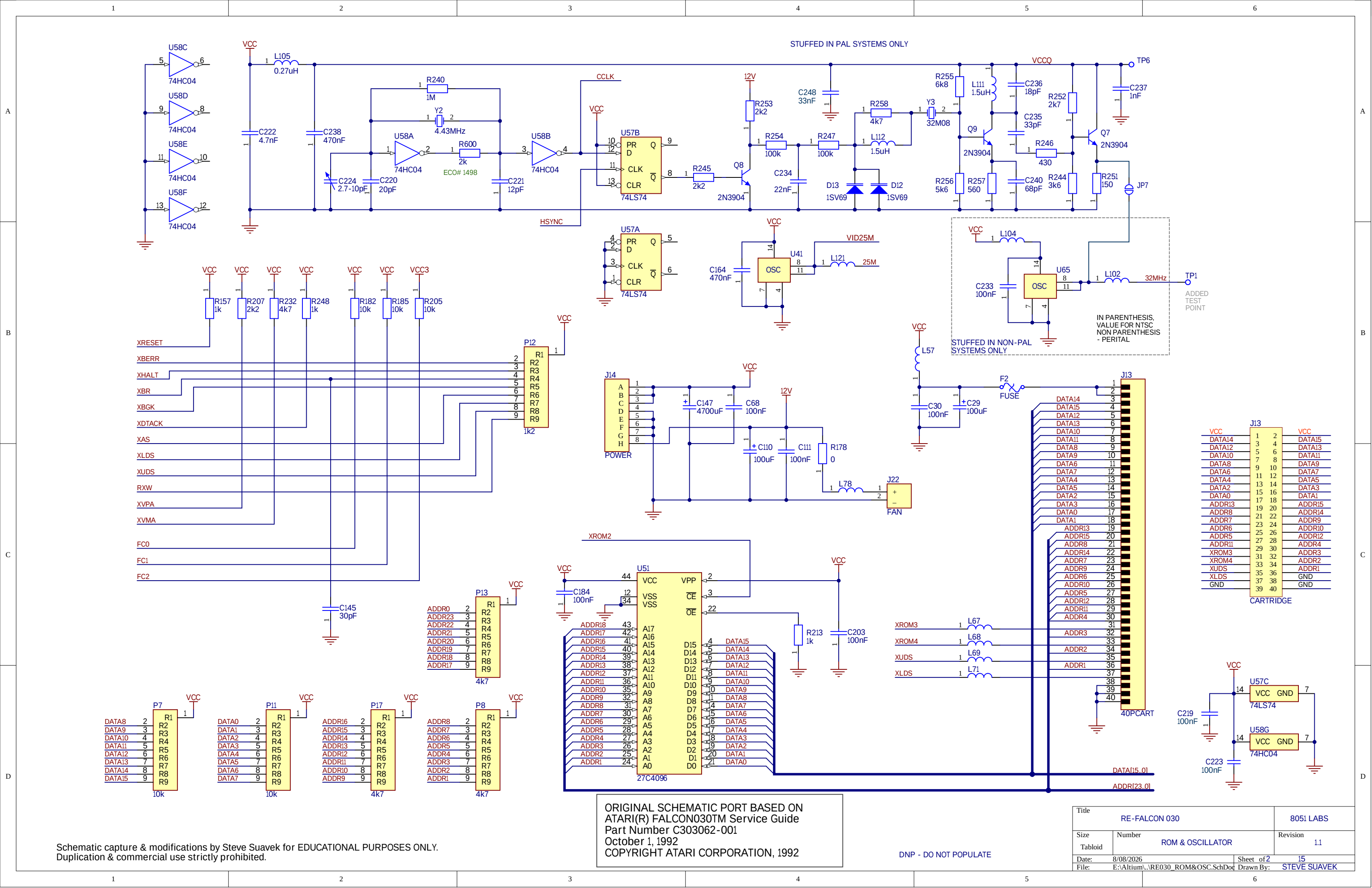
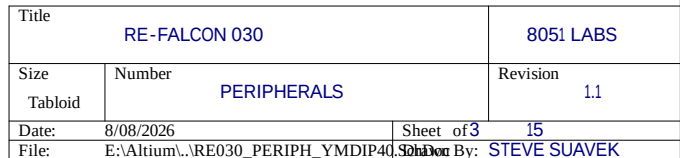


A	1		2	3	4	5	6					
	RE-FALCON 030 REV 1.1 * RELEASE CANDIDATE *						DATE	REV	CHANGE EVENT			
							2025-12-03	1	1st prototype * fixed J6-48 tie to CD (was CDATA0) * J6-38 separated from IO net * VEE separated from VEE2 * fixed C6 & C47 polarity mark * adjusted FDD drive posts holes * added JP8 for DSP CLK flexibility			
B	ECO# 1604 *) 1. DOES NOT AFFECT PCB LAYOUT 2. DOES NOT AFFECT PCB LAYOUT 3. DOES NOT AFFECT PCB LAYOUT 4. C602 ADD TO LAYOUT 1nF 5. C603 ADD TO LAYOUT 1nF 6. C604 ADD TO LAYOUT 1nF 7. C605 ADD TO LAYOUT 1nF 8. C606 ADD TO LAYOUT 1nF 9. C607 ADD TO LAYOUT 1nF 10. C608 ADD TO LAYOUT 1nF 11. C609 ADD TO LAYOUT 1nF 12,13,14 - ADDED TO REPLICA (FOR FCC ONLY) 15,16,17,18 - UNKNOWN PN 50048/49 NOT IN REPLICA 19,20 MECHANICAL AGND BONDING TO CHASSIS - NOT IN PCB ECO# 1600*) ADDED TO SCH & PCB ECO# 1488*) ADDED JP3 ~ JP6 TO SUPPORT ECO# 1498*) ADDED TO SCH & PCB ECO# 1641*) - MARKED IN SCH & BOM FOR DNP - ADDED TO SCH & PCB - ADDED TO SCH & PCB - ADDED TO SCH & PCB - ADDED TO SCH & PCB - ADDED TO SCH & PCB - C91 MIGHT BE TYPO FOR C95, PHYSICAL PCB HAS CAP VALUE TBD - ADDED TO SCH & PCB ECO# 1643*) 11. PRESENT ON SCH & PCB, NOT FULLY CLEAR REQ. C208 IS DECOUPLING CAP, NO RELEVANCE						2026-08-08	1.1	GitHub release candidate			
							The 600 parts are FSM #ECO added. The 700 parts added to Replica with Suavek design changes/improvements.					
	C											
D												
ORIGINAL SCHEMATIC PORT BASED ON ATARI(R) FALCON030TM Service Guide (FSM) Part Number C303062-001 October 1, 1992 COPYRIGHT ATARI CORPORATION, 1992		Disclaimer: This project contains or is based on copyrighted material belonging to its respective owner(s). The author does not claim any ownership or copyright in the original work. Any copyright asserted by the author applies solely to the author's original contributions, including modifications, enhancements, documentation, or other newly created content, and does not extend to the underlying copyrighted material.		ATARI FALCON™ IS A TRADEMARK REGISTERED BY ATARI INTERACTIVE, INC. ORIGINAL DESIGN UNDER COPYRIGHT OF ATARI CORPORATION (C) 1992								
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						TitleTITLE & CHANGE LOG				8051 LABS		
Size		Number		Revision								
Tabloid		6000825		1.1								
Date:		8/08/2026		Sheet		15						
File:		E:\Altium\..RE030_TITLE.SchDoc		Drawn By:		STEVE SUAVEK						
1		2		3		4				5		
										6		



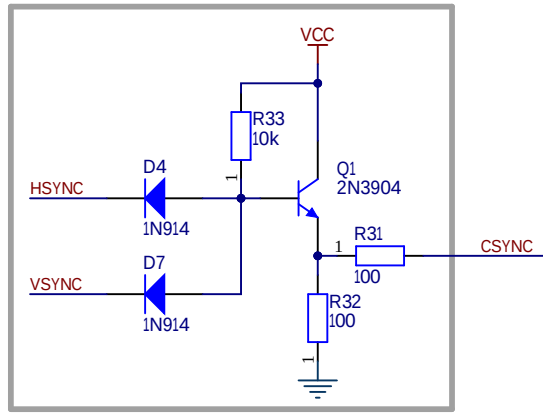
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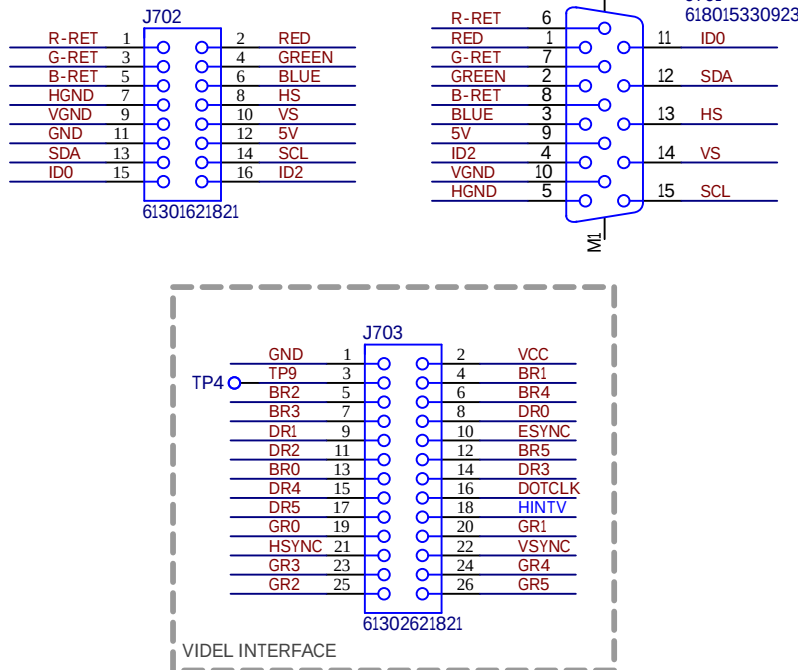
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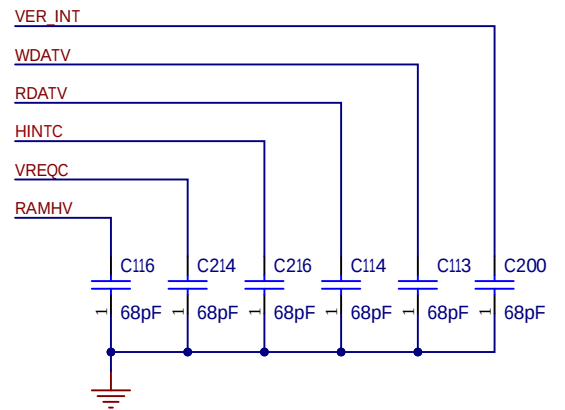
FOR NON MODULATOR SYSTEMS ONLY



REMOVE FOR PAL



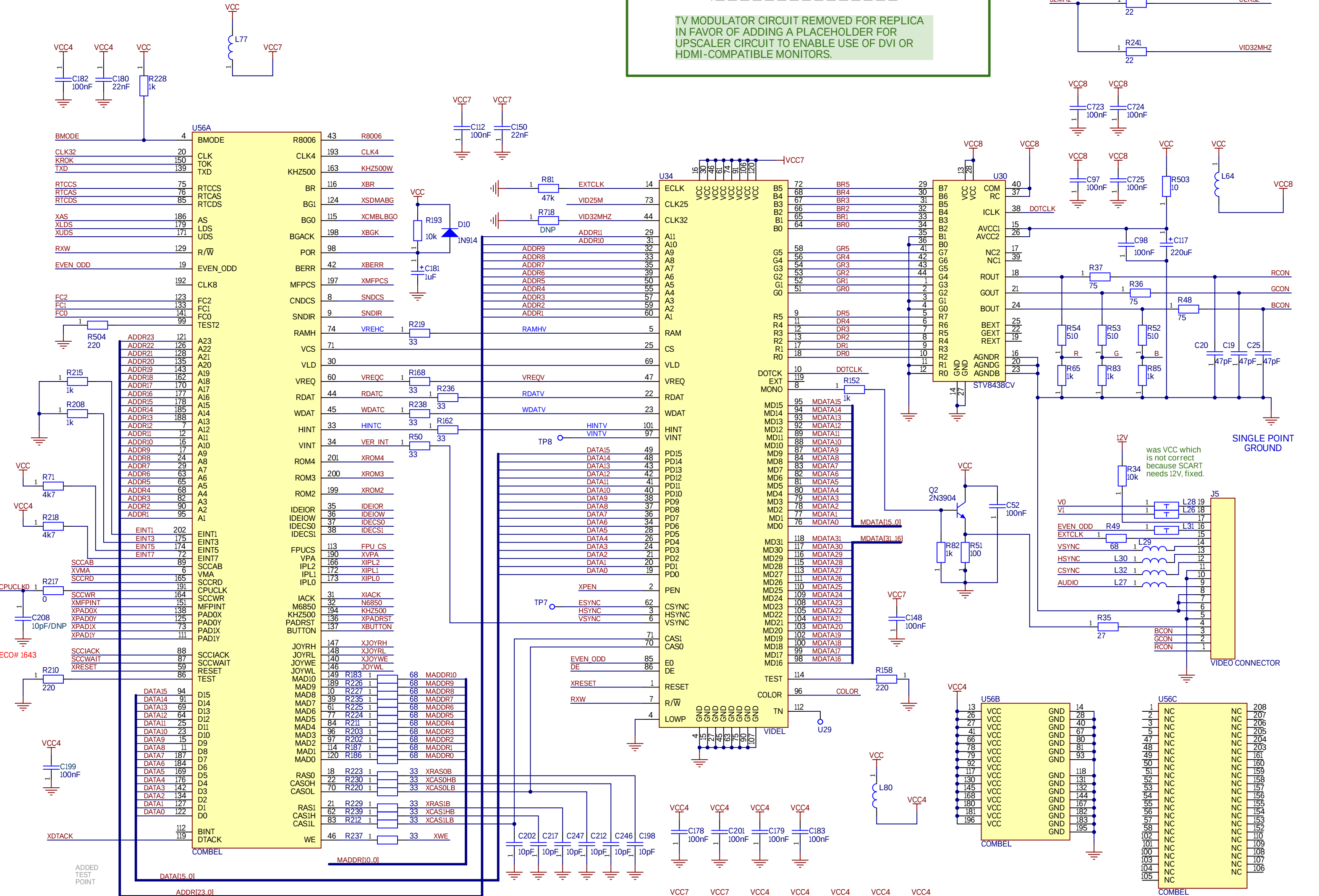
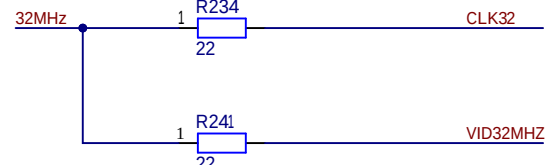
TV MODULATOR CIRCUIT REMOVED FOR REPLICA
IN FAVOR OF ADDING A PLACEHOLDER FOR
UPSCALER CIRCUIT TO ENABLE USE OF DVI OR
HDMI-COMPATIBLE MONITORS.



* - REQUIRES VALUE CHANGE FOR PAL

FOR MODULATOR SYSTEMS ONLY

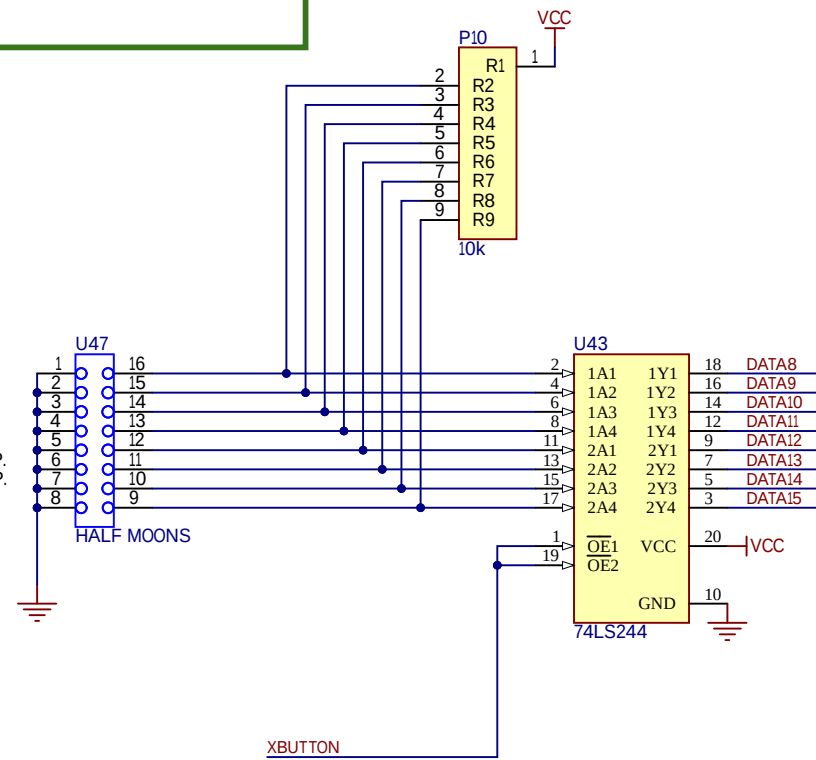
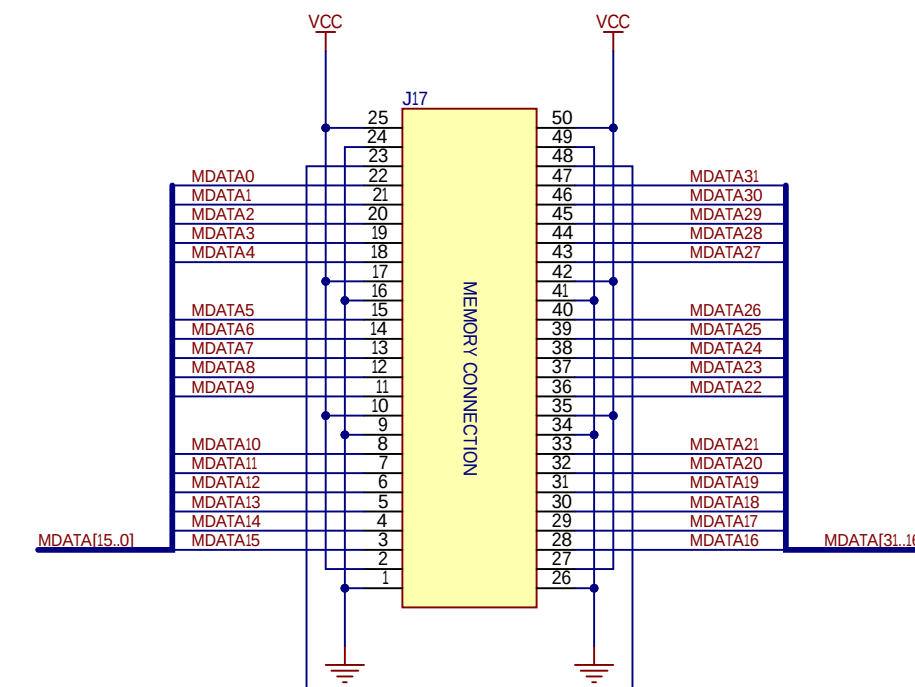
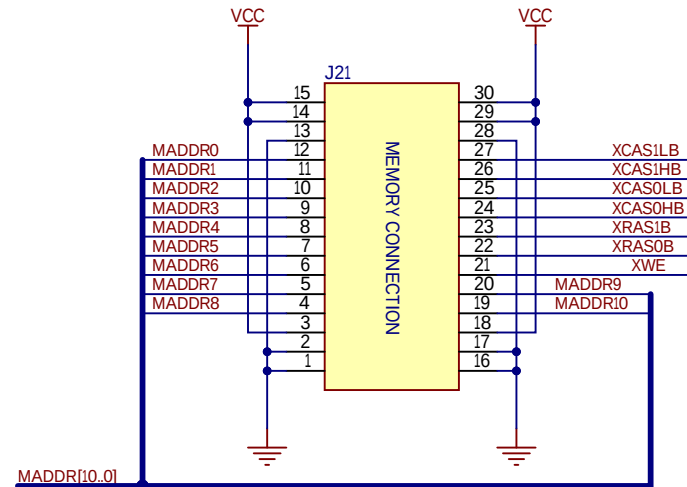
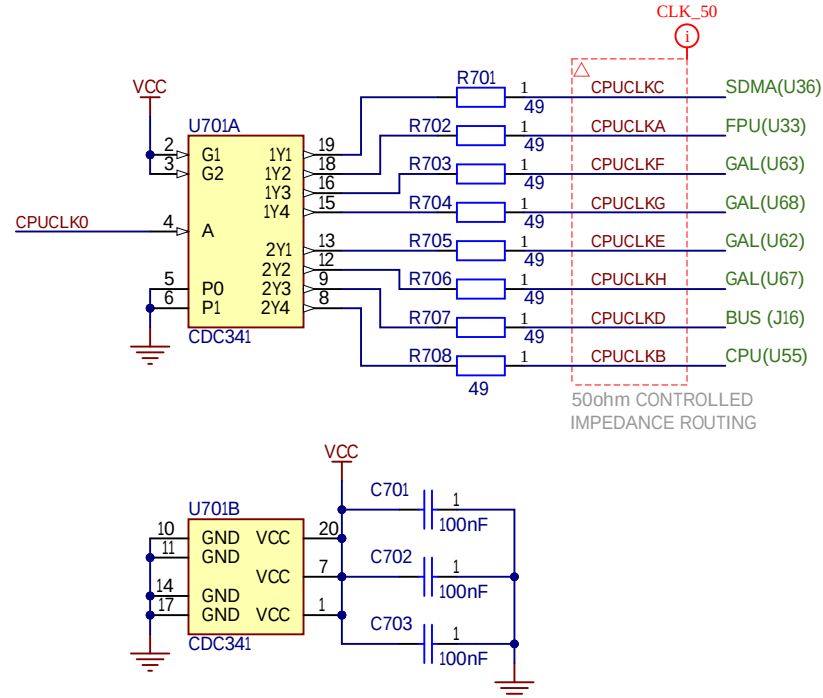
NOTE: JMP2_ST MUST BE PLACED NEAR U35



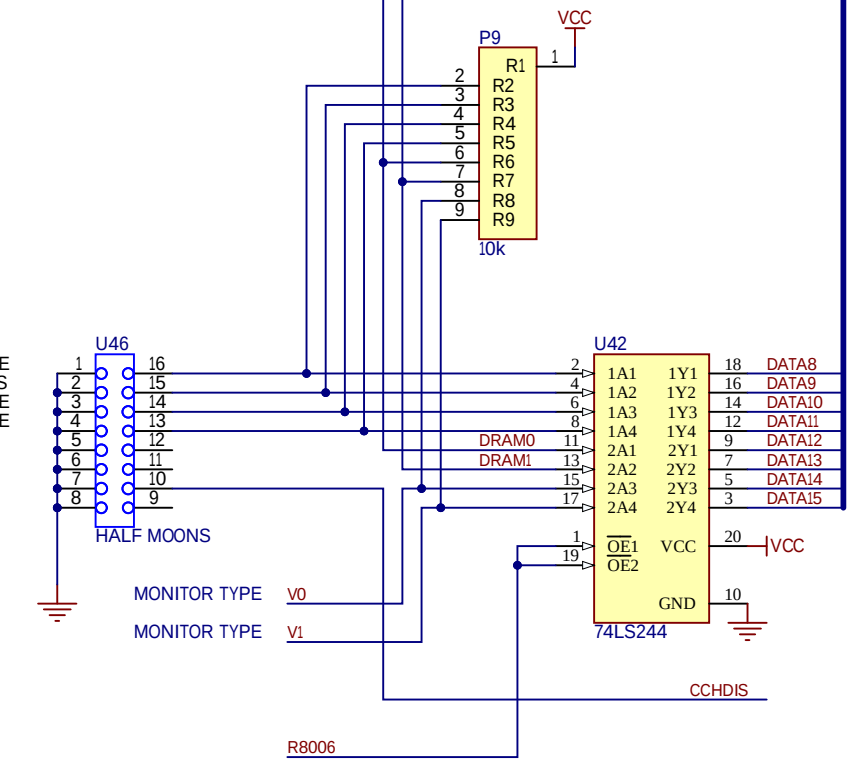
DNP - DO NOT POPULATE

Title			RE-FALCON 030		8051 LABS	
Size	Number		VIDEL & COMBEL		Revision	
	Tabloid				1.1	
Date:	8/08/2026		Sheet of 4		15	
File:	E:\Altium\RE030_VIDEL_NOMODUL\RE030_VIDEL_NOMODUL.DSN		Author		STEVE SUAVEK	

SUAVEK CLOCK DISTRIBUTION CIRCUIT



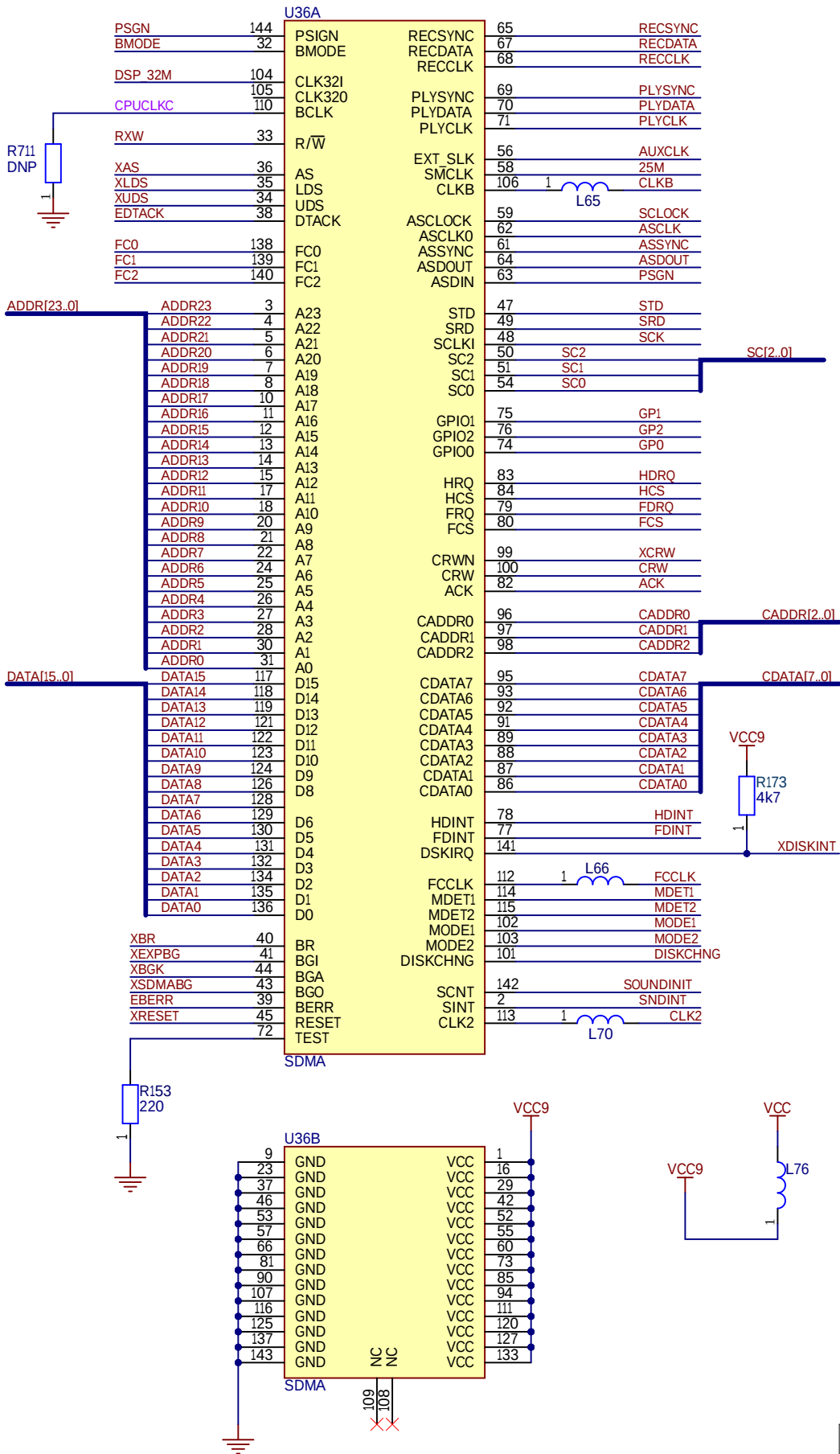
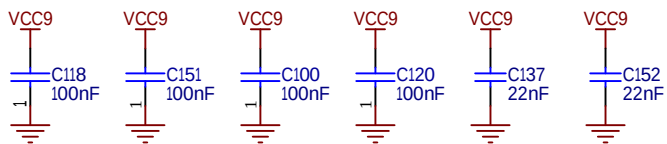
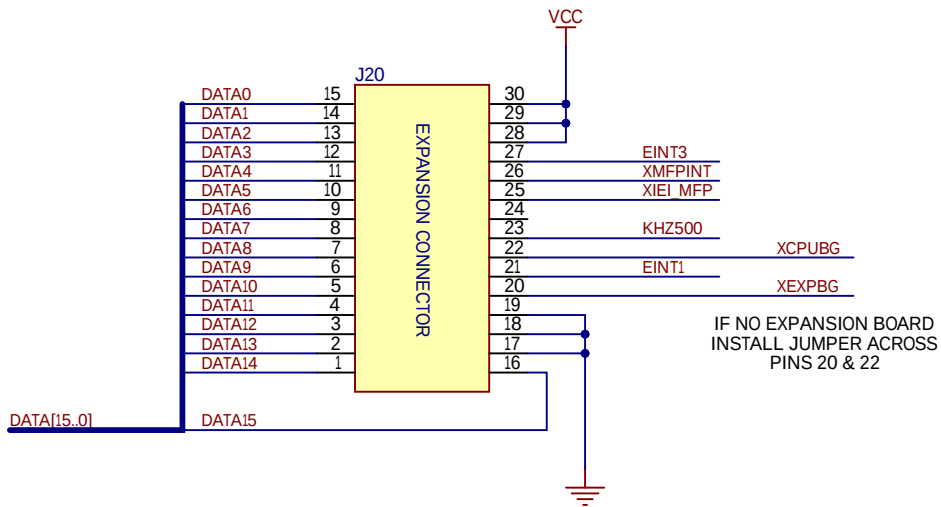
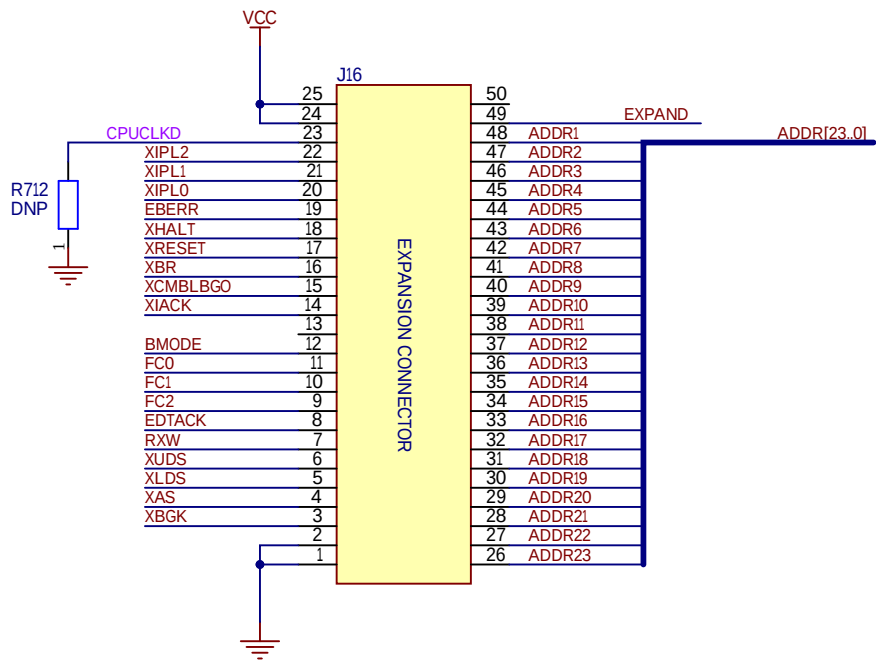
RAM WAIT STATE
16/32 VIDEO BUS
ROM WAIT STATE
ROM WAIT STATE
DRAM TYPE
DRAM TYPE



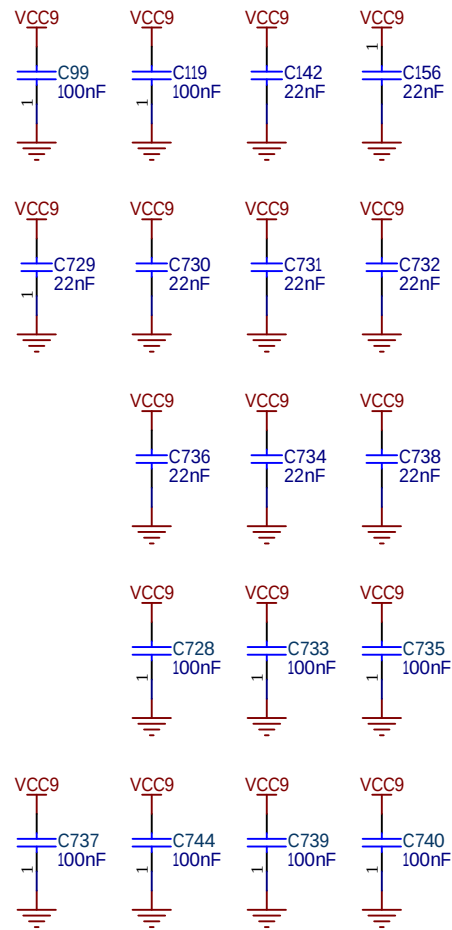
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Part Number C303062-001
October 1, 1992
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Title		RE -FALCON 030	8051 LABS
Size	Number	RAM	Revision
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File:	E:\Altium\...RE030_RAM.SchDoc	Drawn By:	STEVE SUAVEK



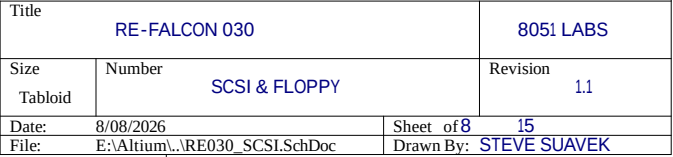
DNP - DO NOT POPULATE



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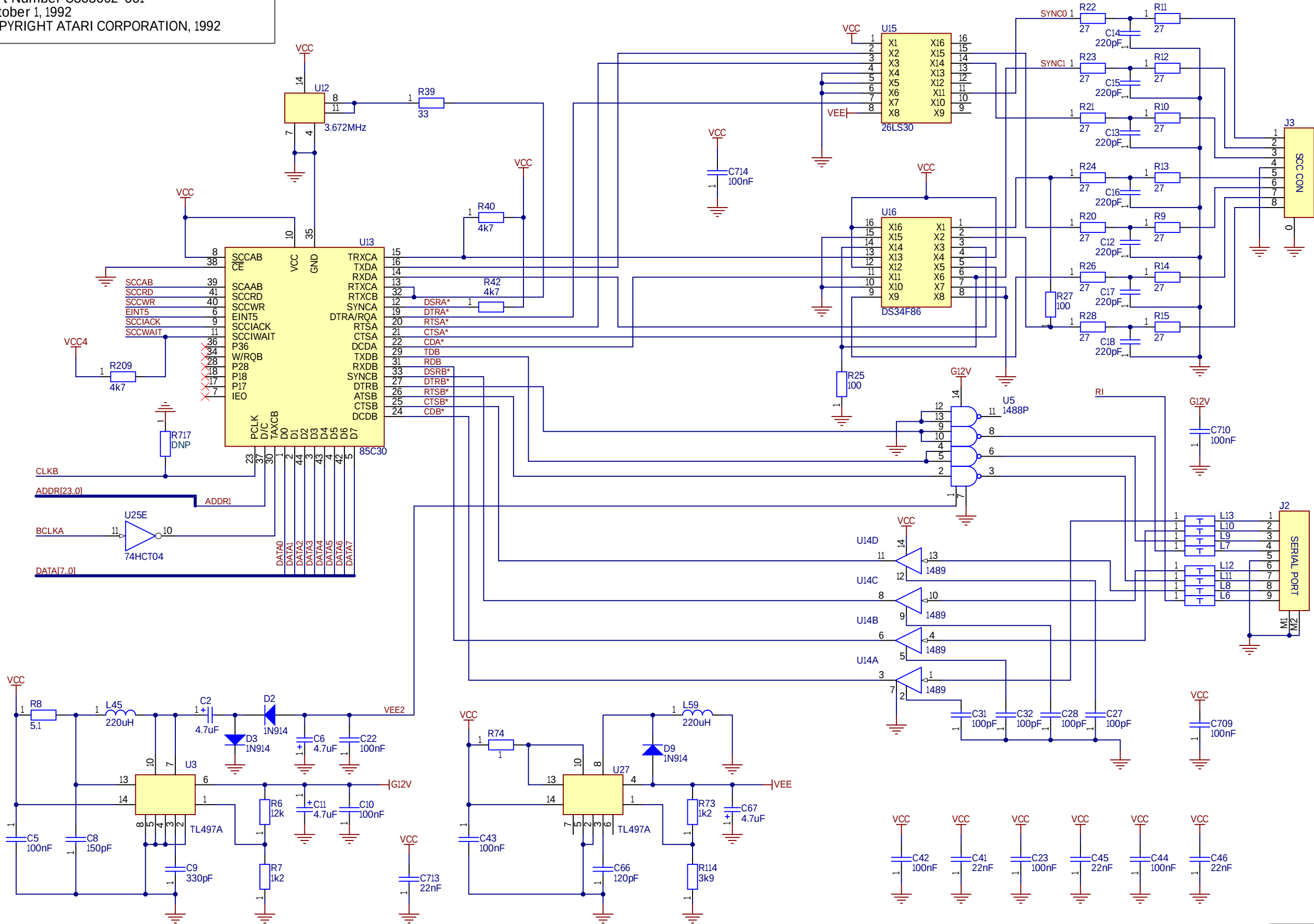
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File:	E:\Altium\...RE030_SDMA.SchDoc	Drawn By:		STEVE SUAVEK	



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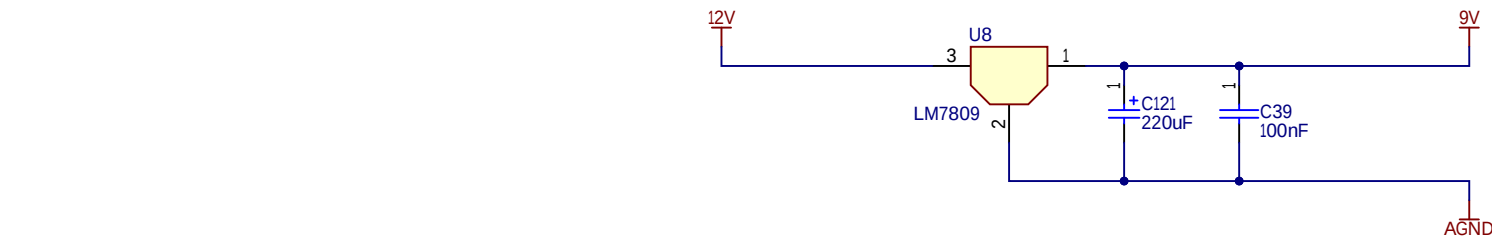


DNP - DO NOT POPULATE

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Size	Number	SERIAL		Revision	
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File:	E:\Altium\...RE030_SERIAL.SchDoc	Drawn By:		STEVE SUAVEK	

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CLOCK TRACES PARTICULARLY SCLOCK, ?MCLK, AUXCLK MUST BE WIDE.
WITH AS FEW TURNS/FEEDTHROUGHTS AS POSSIBLE.

USED CURVED TRACES IF POSSIBLE.

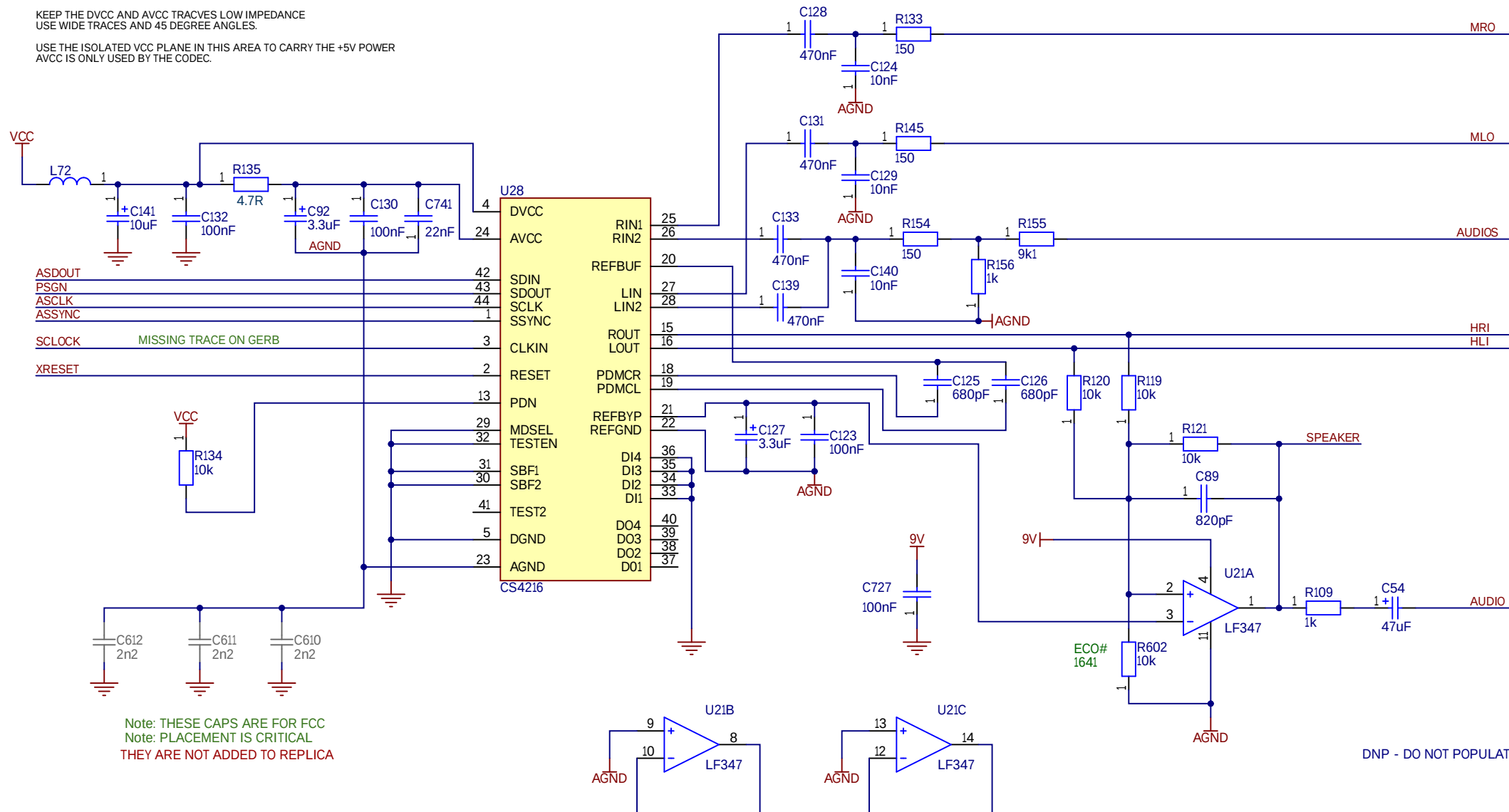
KEEP TRACE WIDTH CONTANT IF NOT THEN TAPER SIGNALS TO AVOID SUDDENT
IMPEDANCE CHANGES.

KEEP COMPONENTS CLOSE TO THE IC'S AND LAY THEM OUT SO THAT THE SIGNAL RUNS
ARE AS STRAIGHT AS POSSIBLE.

USE THE FERRITE BEAD TO BRIDGE THE ISOLATION GAP BETWEEN THE SYSTEM AND
AUDIO VCC AND GND PLANES.

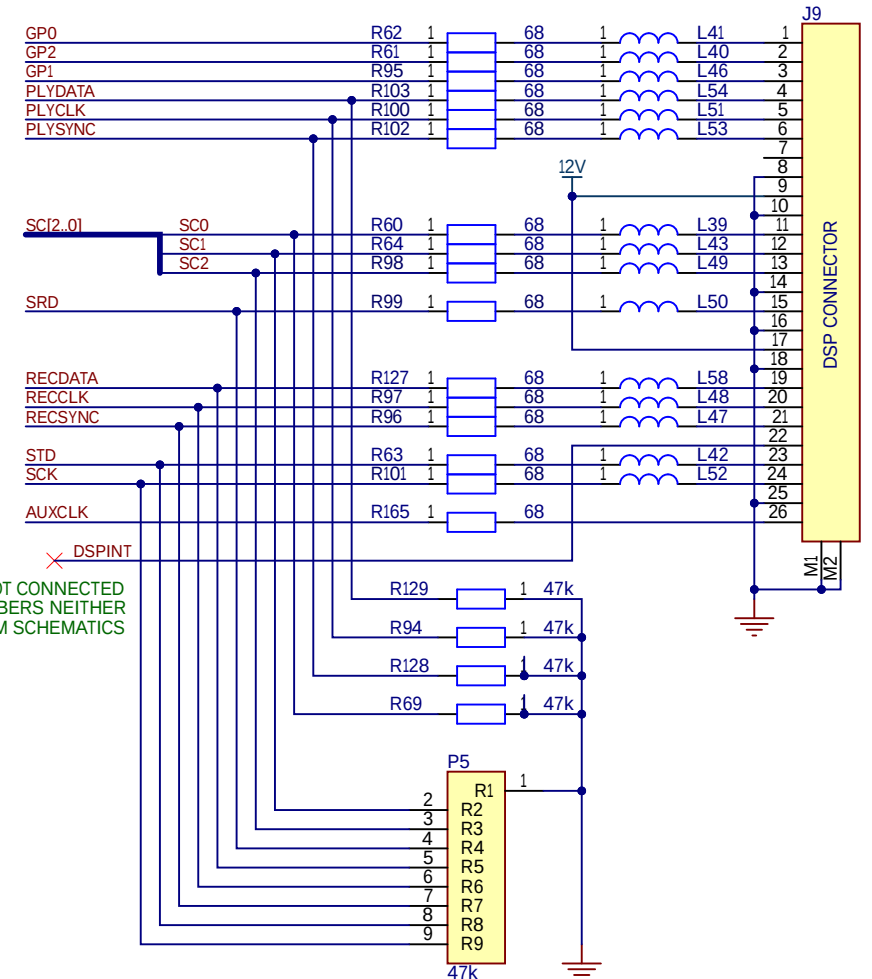
KEEP THE DVCC AND AVCC TRACVES LOW IMPEDANCE
USE WIDE TRACES AND 45 DEGREE ANGLES.

USE THE ISOLATED VCC PLANE IN THIS AREA TO CARRY THE +5V POWER
AVCC IS ONLY USED BY THE CODEC.



Note: THESE CAPS ARE FOR FCC
Note: PLACEMENT IS CRITICAL
THEY ARE NOT ADDED TO REPLICA

LEAVING UNBIASED OP-AMP INPUTS IS BAD DESIGN - FIXED HERE



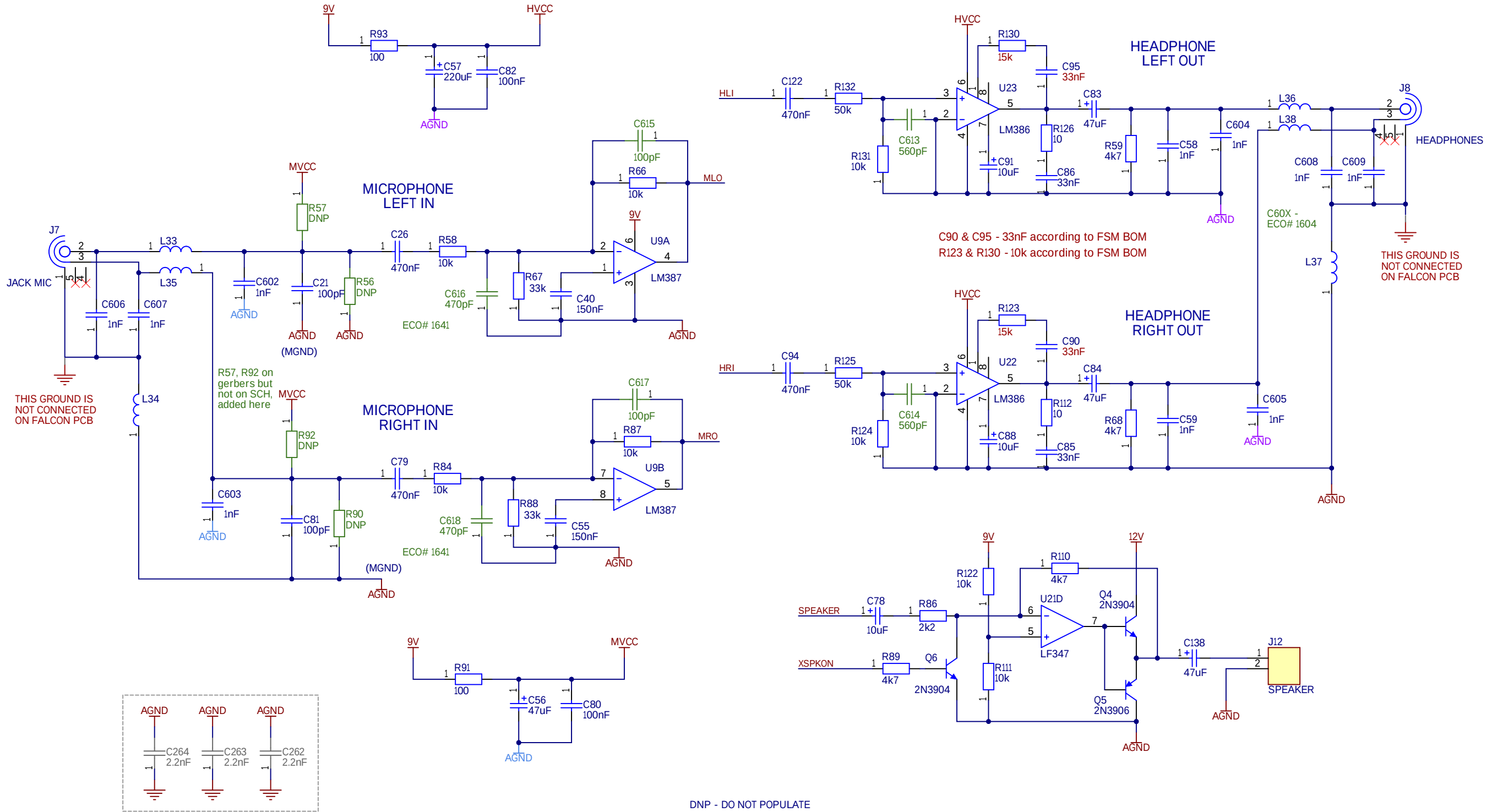
J9-26 NOT CONNECTED
IN UK GERBERS NEITHER
IN SM SCHEMATICS

DNP - DO NOT POPULATE

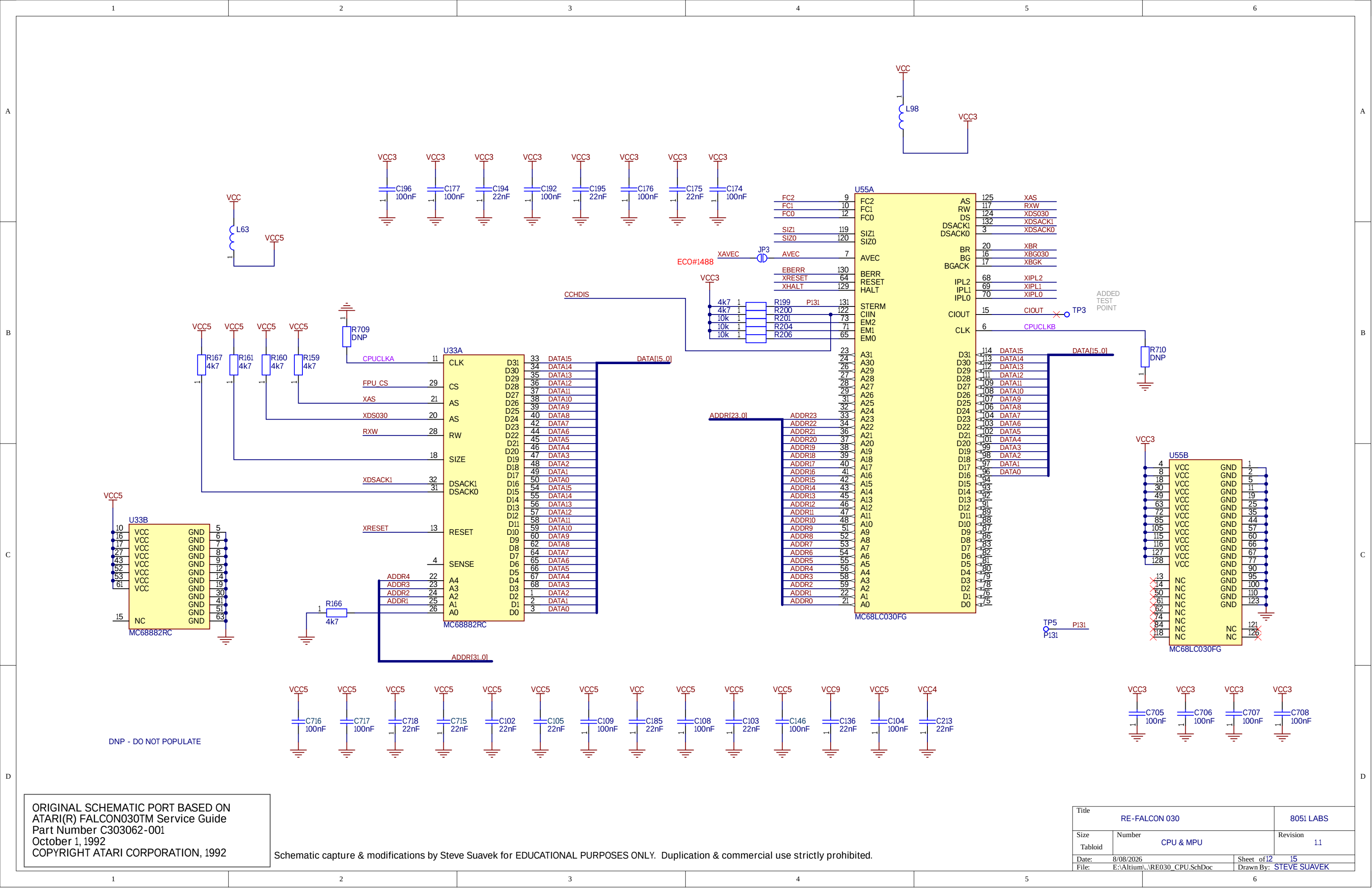
Title		RE -FALCON 030		8051 LABS	
Size	Number	CODEC & SPK AMP		Revision	1.1
Tabloid					
Date:	8/08/2026			Sheet of 9	15
File:	E:\Altium\RE030_CODEC.SchDoc			Drawn By:	STEVE SUAVEK

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Title		RE-FALCON 030		8051 LABS	
Size	Number	CODEC & SPK AMP		Revision	
Tabloid				1.1	
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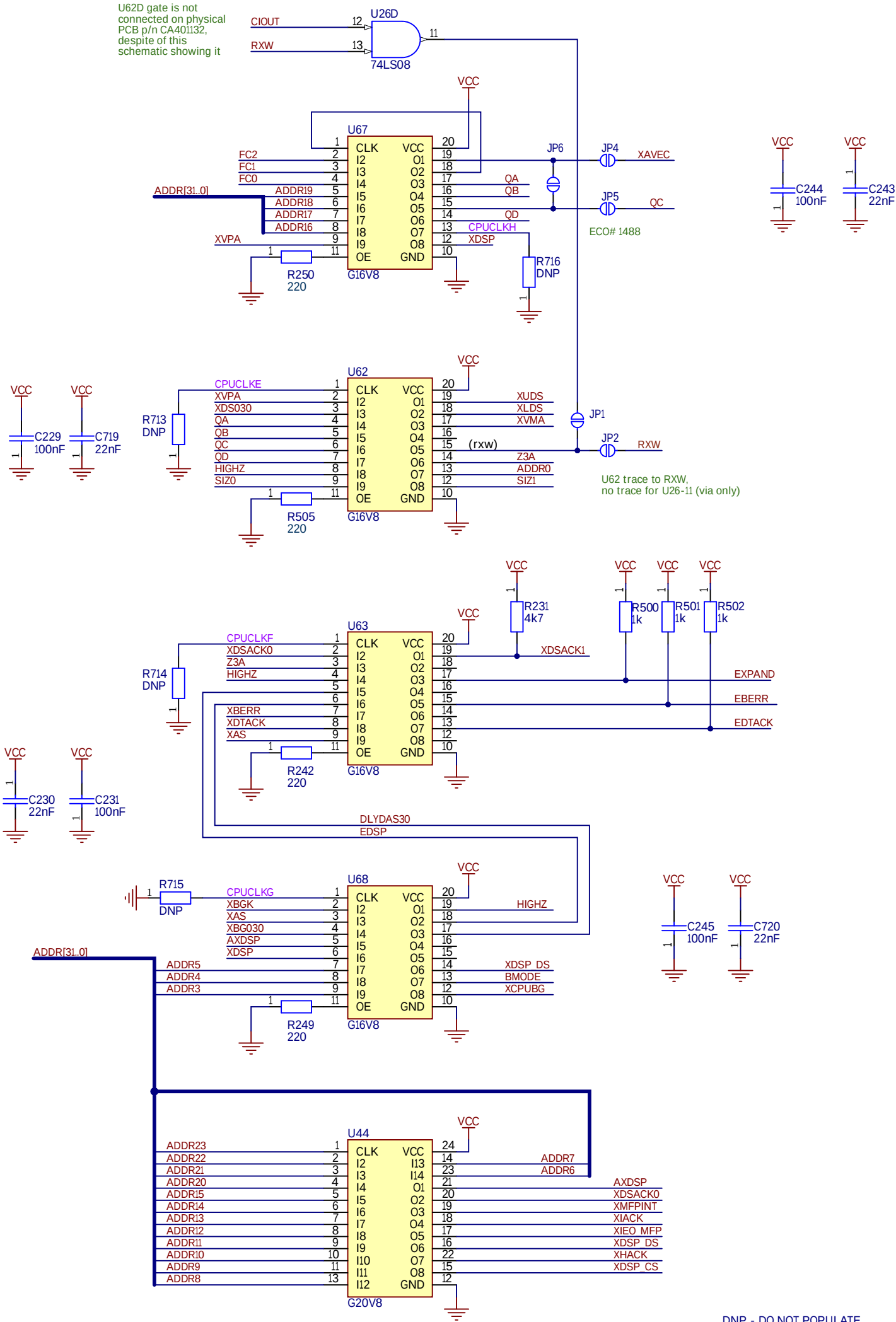


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Title		RE -FALCON 030		8051 LABS	
Size	Number	CPU & MPU		Revision	
Tabloid				1.1	
Date:	8/08/2026	Sheet of 12		15	
File:	E:\Altium\...RE030_CPU.SchDoc	Drawn By:		STEVE SUAVEK	

U62D gate is not connected on physical PCB p/n CA401132, despite of this schematic showing it

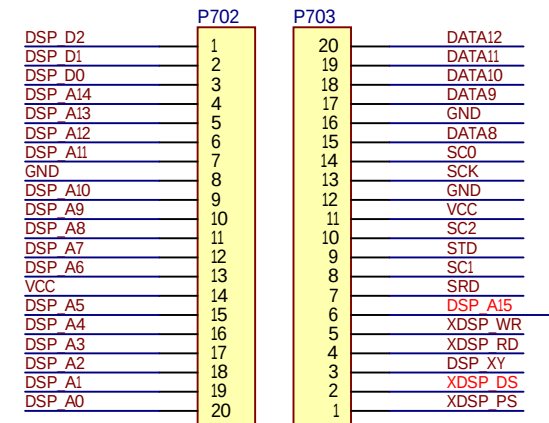
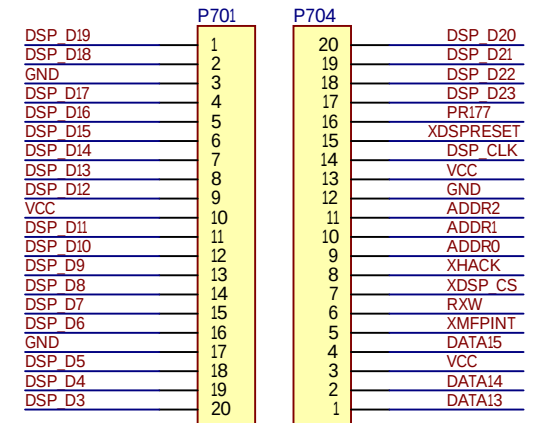


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Title		RE-FALCON 030	8051 LABS
Size	Number	CPU & MPU	Revision
Legal			1.1
Date:	8/08/2026	Sheet of 13	15
File:	E:\Altium\..RE030_GALS.SchDoc	Drawn By:	STEVE SUAVEK

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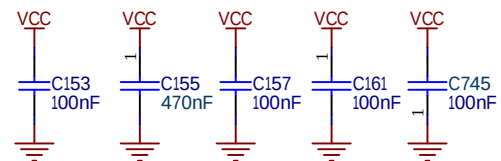
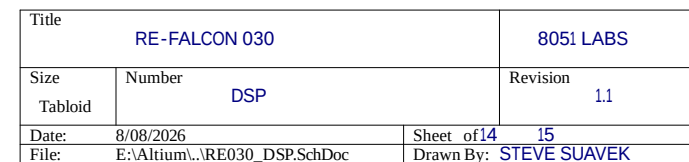
J701~704: POPULATE ONLY IF U38 NOT USED

RED NETS NOT PRESENT
IN ORIGINAL FALCON
DSP BUT USED HERE
FOR FUTURE EXTENSION

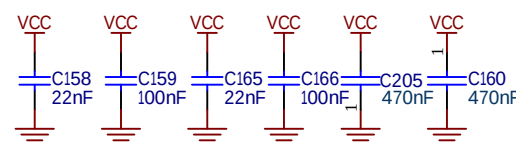
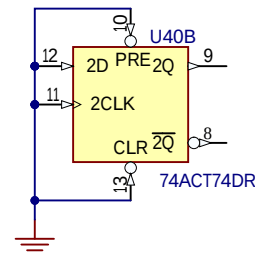
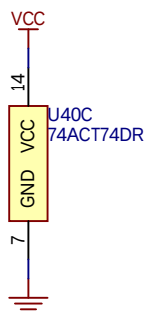
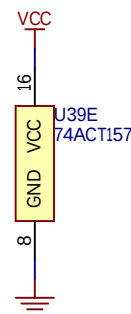
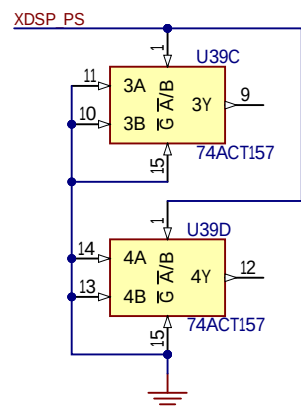
FIDUCIALS FOR PCB ASSEMBLY (NOT BOM)

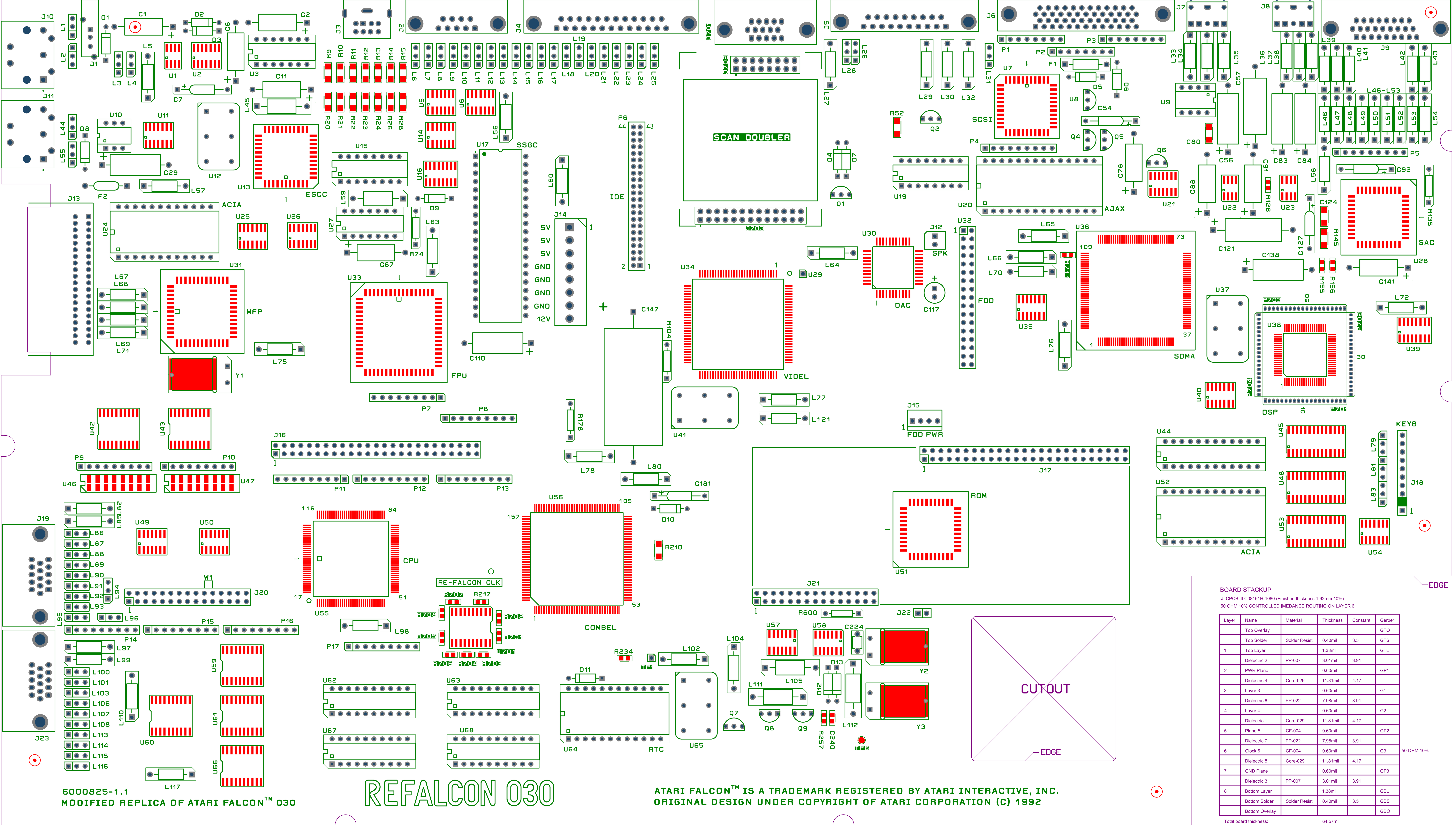


DSP
BOOSTRAP
RESISTOR



DNP - DO NOT POPULATE





6000825-1.1
MODIFIED REPLICA OF ATARI FALCON™ 030

REFALCON 030

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BOARD STACKUP
JLPCB JLC08161H-1080 (Finished thickness 1.62mm 10%)
50 OHM 10% CONTROLLED IMEDANCE ROUTING ON LAYER 6

Layer	Name	Material	Thickness	Constant	Gerber
	Top Overlay				GTO
	Top Solder	Solder Resist	0.40mil	3.5	GTS
1	Top Layer		1.38mil		GTL
	Dielectric 2	PP-007	3.01mil	3.91	
2	PWR Plane		0.60mil		GP1
	Dielectric 4	Core-029	11.81mil	4.17	
3	Layer 3		0.60mil		G1
	Dielectric 6	PP-022	7.98mil	3.91	
4	Layer 4		0.60mil		G2
	Dielectric 1	Core-029	11.81mil	4.17	
5	Plane 5	CF-004	0.60mil		GP2
	Dielectric 7	PP-022	7.98mil	3.91	
6	Clock 6	CF-004	0.60mil		G3
	Dielectric 8	Core-029	11.81mil	4.17	
7	GND Plane		0.60mil		GP3
	Dielectric 3	PP-007	3.01mil	3.91	
8	Bottom Layer		1.38mil		GBL
	Bottom Solder	Solder Resist	0.40mil	3.5	GBS
	Bottom Overlay				GBO

Total board thickness: 64.57mil

EDGE

EDGE

50 OHM 10%

Designator	Name	Quantity	Manufacturer Part Number	Manufacturer	Description	Alt Part or Accessory	Alt Manufacturer
C1	22uF	1	226TTA035M	Cornell	CAP ALUM 22UF 20% 35V AXIAL TH		
C2, C6, C11, C67	4.7uF	4	475TTA050M	Cornell	CAP ALUM 4.7UF 20% 50V AXIAL TH		
C3, C169, C171, C189, C191	10nF	5	C0805C103K5REC AUTO	KEMET	CAP CER 0805 10NF 50V X7R 10%		
C4, C5, C10, C22, C23, C30, C34, C38, C39, C42, C43, C44, C47, C52, C53, C64, C65, C68, C76, C77, C80, C82, C97, C98, C99, C100, C101, C104, C108, C109, C111, C112, C115, C118, C119, C120, C123, C130, C132, C134, C135, C143, C144, C146, C148, C149, C151, C153, C154, C157, C159, C161, C162, C163, C166, C167, C174, C176, C177, C178, C179, C182, C183, C184, C186, C192, C193, C197, C199, C201, C203, C204, C207, C209, C210, C211, C218, C219, C223, C226, C228, C229, C231, C233, C242, C244, C245, C701, C702, C703, C712, C714, C716,	100nF	107	C1206C104K5RAC TU	KEMET	CAP CER 0.1UF 50V X7R 1206		
C7, C181	1uF	2	105TTA050M	Cornell	CAP ALUM 1UF 20% 50V AXIAL TH		
C8	150pF	1	CC0805KRX7R9BB 151	Yageo	CAP CER 150PF 50V X7R 0805		
C9	330pF	1	CC0805KRX7R9BB 331	Yageo	CAP CER 330PF 50V 10%		

Designator	Name	Quantity	Manufacturer Part Number	Manufacturer	Description	Alt Part or Accessory	Alt Manufacturer
C12, C13, C14, C15, C16, C17, C18	220pF	7	CC0805KRX7R9BB221	Yageo	CAP CER220PF50VX7R0805		
C19, C20, C25	47pF	3	CML0805C0G470JT50V	Stackpole	CAP CER30PF50VC0G/NP0 0805		
C21, C27, C28, C31, C32, C33, C81, C615, C617	100pF	9	C0805C101K5GAC TU	KEMET	CAP CER100PF50VC0G/NP0 0805		
C26, C79, C94, C122, C128, C131, C133, C139	470nF	8	C1210C474K5RAC TU	KEMET	CAP CER0.47UF50VX7R1210		
C29, C110	100uF	2	107TTA025M	Cornell	CAP ALUM100UF20%25VAXIAL TH		
C35, C155, C160, C164, C205, C238	470nF	6	CL31B474KBHNN NE	Samsung	CAP CER0.47UF50VX7R1206		
C40, C55	150nF	2	KGM32ER71H154 KU	Kyocera	CAP CER0.15UF50VX7R1210		
C41, C45, C46, C74, C102, C103, C105, C136, C137, C142, C150, C152, C156, C158, C165, C175, C180, C185, C194, C195, C206, C213, C215, C225, C227, C230, C234, C241, C243, C711, C713, C715, C718, C719, C720, C722, C729, C730, C731, C732, C734, C736, C738, C741, C742, C743	22nF	46	C0805C223K5REC AUTO	KEMET	CAP CER0805 22NF50VX7R10%		
C54, C56, C83, C84, C138	47uF	5	476TTA016M	Cornell	CAP ALUM47UF20%16VAXIAL TH		
C57, C121	220uF	2	227TTA025M	Cornell	CAP ALUM220UF20%25VAXIAL TH		
C58, C59, C237, C602, C603, C604, C605, C606, C607, C608, C609	1nF	11	CC0805KRX7R9BB102	Yageo	CAP CER1000PF50VX7R0805		

Designator	Name	Quantity	Manufacturer Part Number	Manufacturer	Description	Alt Part or Accessory	Alt Manufacturer
C63, C113, C114, C116, C187, C200, C214, C216, C240	68pF	9	CC0805JRNPO9BN680	Yageo	CAP CER 68PF 50V C0G/NPO 0805		
C66	120pF	1	C0805C121K5HAC TU	KEMET	CAP CER 120PF 50V X8R 0805		
C78, C88, C91, C141	10uF	4	106TTA100M	Cornell	CAP ALUM 10UF 20% 100V AXIAL TH		
C85, C86, C90, C95, C248	33nF	5	C0805C333K5REC AUTO	KEMET	CAP CER 0805 33NF 50V X7R 10%		
C89, C168, C170, C188, C190	820pF	5	CC0805KRX7R9BB821	Yageo	CAP CER 820PF 50V X7R 0805		
C92, C127	3.3uF	2	335TTA100M	Cornell	CAP ALUM 3.3UF 20% 100V AXIAL TH		
C106, C107, C145	30pF	3	CML0805C0G300JT50V	Stackpole	CAP CER 30PF 50V C0G/NP0 0805		
C117	220uF	1	ECA-1CM221	Panasonic	CAP ALUM 220UF 20% 16V RADIAL TH		
C124, C129, C140	10nF	3	GMC31X7R103K50NT	Cal Chip	CAP CER 10000PF 50V X7R 1206		
C125, C126	680pF	2	CC0805KRX7R9BB681	Yageo	CAP CER 680PF 50V X7R 0805		
C147	4700uF	1	478TTA016M	Cornell	CAP ALUM 4700UF 20% 16V AXIAL TH		
C172, C173, C196, C704, C705, C706, C707, C708, C709, C710, C727	100nF	11	CC0805KRX7R9BB104	Yageo	CAP CER 0.1UF 50V X7R 0805		
C198, C202, C212, C217, C246, C247	10pF	6	CC0805KRNP09BN100	Yageo	CAP CER 10PF 50V C0G/NPO 0805		
C208, R56, R57, R90, R92, R709, R710, R711, R712, R713, R714, R715, R716, R717, R718	10pF/DNP, DNP	15			DO NOT POPULATE		
C220	20pF	1	CC0805JRNPO9BN200	Yageo	CAP CER 20PF 50V C0G/NPO 0805		
C221	12pF	1	C0805C120K5HAC TU	KEMET	CAP CER 0805 10NF 50V X7R 10%		
C222	4.7nF	1	CC0805KRX7R9BB472	Yageo	CAP CER 4700PF 50V X7R 0805		
C224	2.7-10pF	1	GKG10015	EW Electronics	CAP TRIMMER 3-10PF 100V TH		

Designator	Name	Quantity	Manufacturer Part Number	Manufacturer	Description	Alt Part or Accessory	Alt Manufacturer
C232	47nF	1	CL21B473KBCNN NC	Samsung	CAP CER0.047UF50VX7R0805		
C235	33pF	1	CC0805JRNPO9BN330	Yageo	CAP CER CHIP 33PF 50V5%		
C236	18pF	1	CC0805JRNPO9BN180	Yageo	CAP CERA CHIP 18PF 50V		
C613, C614	560pF	2	C0805C561K5HAC TU	KEMET	CAP CER 560PF 50V X8R0805		
C616, C618	470pF	2	C0805C471K5RAC TU	KEMET	CAP CER 470PF 50V X7R0805		
D1, D2, D3, D4, D7, D8, D9, D10, D11	1N914	9	1N914TR	OnSemi	DIODE STANDARD 100V200MA DO35		
D5, D6	1N4001	2	1N4001	Diotec	DIODE STANDARD 50V1A DO204AC		
D12, D13	1SV69	2	1SV69	OnSemi	SILICON EPITAXIAL PLANAR VARACTOR	BB505B	Siemens
F1, F2	FUSE	2	0251.500MXL	Littlefuse	FUSE BRD MT 500MA 125VAC/VDC AXL		
J1	RES	1	C070119	Atari	RESET PUSHBUTTON	SPJ12U	
J2	SERIAL PORT	1	618009231221	Würth	CONN D-SUB PLUG 9POS R/A SOLDER		
J3	SCC CON	1	KMDGX-8S-AS	Kycon	CONN RCPT FEMALE MINI DIN 8P SLDR		
J4	PARALLEL PORT	1	618025231121	Würth	CONN D-SUB RCPT 25POS R/A SOLDER		
J5	VIDEO CONNECTOR	1	C302823	Atari	CONN D SUB 19P MALE		
J6	SCSI CONNECTOR	1	C302822	Atari	SCSI D SUB 50HD		
J7, J8	JACK MIC, HEADPHONES	2	SJL-3525N	Same Sky	CONN JACK STEREO 3.5MM R/A		
J9	DSP CONNECTOR	1	L17HTHAS4F1C	Amphenol	CONN D-SUB HD RCPT 26P R/A SLDR	L77HDA26SD1CH 4F	Amphenol
J10, J11	MIDI OUT, MIDI IN	2	KCDX-5S-N2	Kycon	CONN RCPT FEMALE MINI DIN 5P SLDR		
J12, J22	SPEAKER, FAN	2	IL-G-2P-S3T2-SA	JAE Electronics	CONN HEADER VERT 2POS 2.5MM	IL-G-2S-S3C2-SA	IL-G-C2-SC-10000
J13	40PCART	1	C070129	Atari	EDGE CONNECTOR 40PIN		
J14	POWER	1	0009652088	Molex	CONN HEADER VERT 8POS 3.96MM	0009501081	0008701031
J15	Header 4	1	LHA-04-TS	Adam Tech	CONN HEADER VERT 4POS 2.54MM	MTA-04	MTA-C-R
J16, J17	EXPANSION CONNECTOR, MEMORY CONNECTION	2	67997-250HLF	Amphenol	CONN HEADER VERT 50POS 2.54MM	HTSW-125-07-G-D	Samtec
J18	KEYBOARD	1	22032101	Molex	CONN HEADER VERT 10POS 2.54MM	remove pin 2	
J19, J23	JOY-B, JOY-A	2	17EBH015SAA110	Amphenol	CONN D-SUB HD RCPT 15P R/A SLDR		

Designator	Name	Quantity	Manufacturer Part Number	Manufacturer	Description	Alt Part or Accessory	Alt Manufacturer
J0, J1	EXPANSION CONNECTOR, MEMORY CONNECTION	2	67997-232HLF	Amphenol	CONN HEADER VERT 32POS 2.54MM	HTSW-115-07-S-D	Samtec
J701	618015330923	1	618015330923	Amphenol	CONN D-SUB HD RCPT 15PR/ASLDR		
J702	61301621821	1	61301621821	Würth Elektronik	WR-PHD 2.54 MM DUAL SOCKET HEADE		
J703	61302621821	1	61302621821	Würth Elektronik	WR-PHD 2.54 MM DUAL SOCKET HEADE		
L1, L2, L3, L4, L6, L7, L8, L9, L10, L11, L12, L13, L14, L15, L16, L17, L18, L19, L20, L21, L22, L23, L24, L25, L26, L28, L31, L44, L55, L79, L81, L83, L86, L87, L88, L89, L90, L91, L92, L93, L94, L95, L96, L100, L101, L103, L106, L107, L108, L113, L114, L115, L116	RC FILTER, [NoValue]	53	DSS1NB32A102Q91A	Murata	FILTER LC(T) 1000PF TH		
L5, L27, L29, L30, L32, L33, L34, L35, L36, L37, L38, L39, L40, L41, L42, L43, L46, L47, L48, L49, L50, L51, L52, L53, L54, L56, L57, L58, L60, L63, L64, L65, L66, L67, L68, L69, L70, L71, L72, L75, L76, L77, L78, L80, L82, L85, L97, L98, L99, L102, L104, L110, L117, L121	FB	54	74276041	Wurth	FERRITE BEAD 89 OHM AXIAL 1LN		
L45, L59	220uH	2	77F221J-TR-RC	Bourns Inc.	FIXED IND 220UH 155MA 5 OHM TH		
L105	0.27uH	1	78FR27K-RC	Bourns Inc.	FIXED IND 270NH 1.11A 85 MOHM TH		
L111, L112	1.5uH	2	5300-03-RC	Bourns Inc.	FIXED IND 1.5UH 3.1A 20 MOHM TH		

Designator	Name	Quantity	Manufacturer Part Number	Manufacturer	Description	Alt Part or Accessory	Alt Manufacturer
P1, P2, P3	220/330	3	4608X-104-221/331L	Bourns	RES NETWORK 12 RES MULT OHM 8SIP		
P4	1k	1	4609X-101-102LF	Bourns	RES ARRAY 8 RES 1K OHM 9SIP		
P5	47k	1	4609X-101-473LF	Bourns	RES ARRAY 8 RES 47K OHM 9SIP		
P6	IDE INTERFACE	1	SBH21-NBPN-D22-ST-BK	Sullins	CONN HEADER VERT 44POS 2MM	62004421121	Würth
P7, P9, P10, P11	10k	4	4609X-101-103LF	Bourns	RES ARRAY 8 RES 10K OHM 9SIP		
P8, P13, P14, P15, P16, P17	4k7	6	4609X-101-472LF	Bourns	RES ARRAY 8 RES 4.7K OHM 9SIP		
P12	1k2	1	4609X-101-122LF	Bourns	RES ARRAY 8 RES 1.2K OHM 9SIP		
P701, P702, P703, P704	Header 20	4	M50-3532042	Harwin	CONN HEADER VERT 20POS 1.27MM	M50-3032042	DNP (optional), Sockets
Q1, Q2, Q4, Q6, Q7, Q8, Q9	2N3904	7	2N3904	Diotec Semi	TRANS NPN 40V 0.2A TO-92		
Q5	2N3906	1	2N3906	Diotec Semi	TRANS PNP 40V 0.2A TO-92		
R1, R4, R18, R19, R41, R113, R158, R242, R249, R250, R504, R505	220	12	RT1206FRE07220 RL	Yageo	RES SMD 220 OHM 1% 1/4W 1206		
R2, R5, R33, R34, R44, R55, R58, R66, R84, R87, R111, R119, R120, R121, R122, R124, R131, R134, R177, R179, R182, R185, R188, R190, R192, R193, R196, R197, R198, R201, R204, R205, R206, R602	10k	34	RT0805FRE0710KL	Yageo	RES SMD 10K OHM 1% 1/8W 0805		
R3, R6	12k	2	RT0805FRE0712KL	Yageo	RES SMD 12K OHM 1% 1/8W 0805		
R7, R73	1k2	2	RT0805FRE071K2L	Yageo	RES SMD 1.2K OHM 1% 1/8W 0805		
R8	5.1	1	RT0805FRE075RL	Yageo	RES SMD 5.1 OHM 1% 1/8W 0805		

Designator	Name	Quantity	Manufacturer Part Number	Manufacturer	Description	Alt Part or Accessory	Alt Manufacturer
R9, R10, R11, R12, R13, R14, R15, R20, R21, R22, R23, R24, R26, R28, R35	27	15	RC1206FR-0727RL	Yageo	RES27 OHM1%1/4W1206		
R16, R17, R38, R40, R42, R59, R68, R71, R72, R89, R110, R147, R159, R160, R161, R166, R167, R173, R176, R180, R181, R194, R199, R200, R209, R214, R218, R231, R232, R258	4k7	30	RT0805FRE074K7L	Yageo	RES SMD 4.7K OHM1%1/8W0805		
R25, R27, R31, R32, R51, R91, R93	100	7	RC1206FR-07100RL	Yageo	RES100 OHM1%1/4W1206		
R29, R148	3k3	2	RT0805FRE073K3L	Yageo	RES SMD 3.3K OHM1%1/8W0805		
R36, R37, R48	75	3	RC1206FR-0775RL	YAGEO	RES75 OHM1%1/4W1206		
R39, R50, R162, R168, R212, R219, R220, R223, R229, R230, R233, R236, R237, R238, R239	33	15	RC0805FR-0733RL	Yageo	RES33 OHM1%1/8W0805		
R43, R114	3k9	2	RT0805FRE073K9L	Yageo	RES SMD 3.9K OHM1%1/8W0805		
R49, R60, R61, R62, R63, R64, R95, R96, R97, R98, R99, R100, R101, R102, R103, R127, R165, R183, R186, R187, R202, R203, R211, R224, R225, R226, R227, R235	68	28	RC0805FR-0768RL	Yageo	RES68 OHM1%1/8W0805		
R52, R53, R54	510	3	RC1206FR-07510RL	YAGEO	RES510 OHM1%1/4W1206		
R65, R83, R85	1k	3	RC1206FR-071KL	YAGEO	RES1K OHM1%1/4W1206		

Designator	Name	Quantity	Manufacturer Part Number	Manufacturer	Description	Alt Part or Accessory	Alt Manufacturer
R67, R88	33k	2	RC0805FR-0733KL	Yageo	RES33KOHM1%1/8W0805		
R69, R81, R94, R128, R129	47k	5	RC0805FR-0747KL	Yageo	RES47KOHM1%1/8W0805		
R74	1	1	CFR-25JT-52-1R	Yageo	RES1 OHM5%1/4WAXIAL	CF1/4CT52R1R0J	Koa
R75, R76, R77, R78, R115, R136, R137, R138, R139, R140, R141, R142, R143, R149, R150, R151	56	16	RC0805FR-1356RL	Yageo	RES56 OHM1%1/8W0805		
R82, R109, R118, R144, R146, R152, R157, R163, R169, R174, R175, R184, R208, R213, R215, R228, R243, R248, R500, R501, R502	1k	21	RT0805FRE071KL	Yageo	RES SMD 1K OHM 1% 1/8W0805		
R86, R207, R245, R253	2k2	4	RT0805FRE132K2L	Yageo	RES2.2KOHM1%1/8W0805		
R104, R178	0	2	CFR-25JR-52-0R	Yageo	RES0 OHM5%1/4WAXIAL		
R112	10	1	RT0805FRD0710RL	Yageo	RES SMD 10 OHM 1% 1/8W0805		
R123, R130	15k	2	RT0805FRE0715KL	Yageo	RES SMD 15K OHM 1% 1/8W0805		
R125, R132	50k	2	RT0805FRE0749K9L	Yageo	RES SMD 49.9K OHM 1% 1/8W0805		
R126	10	1	RMCF0805FT10R0	Staxckpole	RES10 OHM1%1/8W0805		
R133, R145, R154, R251	150	4	RC1206FR-07150RL	YAGEO	RES150 OHM1%1/4W1206		
R135	4.7R	1	CFR-25JR-52-4K7	Yageo	RES4.7KOHM5%1/4WAXIAL		
R153	220	1	RT0805FRE07220RL	Yageo	RES SMD 220 OHM 1% 1/8W0805		
R155	9k1	1	CRCW08059K10FKEA	Vishay	RES SMD 9.1K OHM 1% 1/8W0805		
R156	1k	1	RMCF0805FT1K00	Koa	RES1KOHM1%1/8W0805		
R164, R170, R171, R172	47	4	RC1206FR-0747RL	YAGEO	RES47 OHM1%1/4W1206		
R189, R191, R195, R240	1M	4	RT0805FRD071ML	Yageo	RES SMD 1M OHM 1% 1/8W0805		

Designator	Name	Quantity	Manufacturer Part Number	Manufacturer	Description	Alt Part or Accessory	Alt Manufacturer
R210	220	1	RC1206FR-07220RL	Yageo	RES 220 OHM 1% 1/4W 1206		
R217	0	1	TLRZ2ATTD	Koa	RES 0 OHM JUMPER 0805		
R234	22	1	CRCW080522R0FKEA	Vishay	RES SMD 22 OHM 1% 1/8W 0805		
R241	22	1	RT0805FRE0722RL	Yageo	RES SMD 22 OHM 1% 1/8W 0805		
R244	3k6	1	RC0805FR-073K6L	Yageo	RES 3.6K OHM 1% 1/8W 0805		
R246	430	1	RT0805FRE07430RL	Yageo	RES SMD 430 OHM 1% 1/8W 0805		
R247, R254	100k	2	RT0805FRD07100KL	Yageo	RES SMD 100K OHM 1% 1/8W 0805		
R252	2k7	1	RT0805FRE072K7L	Yageo	RES SMD 2.7K OHM 1% 1/8W 0805		
R255	6k8	1	RT0805FRE076K8L	Yageo	RES SMD 6.8K OHM 1% 1/8W 0805		
R256	5k6	1	RC0805FR-135K6L	Yageo	RES 5.6K OHM 1% 1/8W 0805		
R257	560	1	CRCW0805562RFKEA	Vishay	RES SMD 562 OHM 1% 1/8W 0805		
R503	10	1	RC1206FR-0710RL	YAGEO	RES 10 OHM 1% 1/4W 1206		
R600	2k	1	CFR-25JR-52-2K	Yageo	RES 2K OHM 5% 1/4W AXIAL		
R701, R702, R703, R704, R705, R706, R707, R708	49	8	CRCW080549R9FKEA	Vishay	RES SMD 49.9 OHM 1% 1/8W 0805	may need adjuustment	
U1	LM555	1	LM555CM/NOPB	Texas Instruments	IC OSC SNGL TIMER 100KHZ 8-SOIC		
U2, U11, U35, U54	7406	4	SN74LS06DR	Texas Instruments	IC INVERTER 6CH 1-INP 14SOIC		
U3, U27	TL497A	2	TL497A	Texas Instruments	IC REG BUCK BST ADJ 500MA 14PDIP	A14-LC-TT	Assmann
U5	1488P	1	MAX1488ECSD+	Analog Device	IC TRANSCEIVER 4/0 14SOIC		
U6, U14	1489	2	MAX1489EESD+T	Analog Device	IC TRANSCEIVER 0/4 14SOIC		
U7	L53C80JC2	1	L53C80	3M	SCSI BUS CONTROLLER	8444-21B1-RK-TP	3M
U8	LM7809	1	78L09	WS	LINEAR VOLTAGE REGULATOR TO-92	LM78L09ACZ	Texas Instruments
U9	LM387	1	LM387N	National Semiconductor	IC AUDIO AMP AB DIP8	A08-LC-TT	Assmann
U10	PC900	1	PC900	Sharp	OPTOISOLTR 5.3KV OPEN COLL 6-DIP	H11L3	Harris
U12	3.672MHz	1	3.6720MHz	MPC	OSCILLATOR 3.6720 MHZ	AR14-HZL/01-TT	Assmann
U13	85C30	1	Z80C30	Zilog	IC INTERFACE SPECIALIZED 44PLCC	8444-21B1-RK-TP	3M
U15	26LS30	1	AM26LS30	Motorola	IC TRANSCEIVER 4/0 DIP16	A16-LC-TT	Assmann

Designator	Name	Quantity	Manufacturer Part Number	Manufacturer	Description	Alt Part or Accessory	Alt Manufacturer
U16	DS34F86	1	DS34C86TMX/NOPB	Texas Instruments	IC TRANSCEIVER0/4 16SOIC		
U17	YM2149	1	YM2149F	Yamaha	SOUND GENERATOR	A40-LC-TT	Assmann
U19	26LS31	1	AM26LS31CN	Texas Instruments	IC TRANSCEIVER4/0 DIP16	A16-LC-TT	Assmann
U20	AJAX	1	C302096	Atari	ATARI FLOPPY CONTROLLER EXT.	A28-LC-TT	Assmann
U21	LF347	1	LF347DR	Texas Instruments	IC OPAMP JFET 4 CIRCUIT 14SOIC		
U22, U23	LM386	2	LM386MX-1	Texas Instruments	IC AUDIO AMP AB 8-SOIC		
U24	MC6850	1	MC6850	Atari	ASYN COMMUNICATION INTERFACE	A24-LC-TT	Assmann
U25	74HCT04	1	SN74HCT04DR	Texas Instruments	IC INVERTER6CH 1-INP14SOIC		
U26	74LS08	1	SN74LS08DR	Texas Instruments	IC GATE AND 4CH 2-INP14SOIC		
U28	CS4216	1	CS4216	Cirrus Logic	CODEC AUDIO	8444-21B1-RK-TP	3M
U30	STV8438CV	1	STV8438CV	ST Microelectronics	VIDEO DAC		
U31	MC68901	1	MC68901	Motorola	MULTI FUNCTION PERIPHERAL	8452-21B1-RK-TP	3M
U32	FLOPPY CABLE	1	61303421121	Würth	CONN HEADER VERT 34POS 2.54MM		
U33	MC68882RC	1	MC68882RC	Motorola	FLOATING POINT PROCESSOR	8468-21B1-RK-TP	3M
U34	VIDEL	1	C302829	Atari	VIDEL CUSTOM CHIPSET		
U36	SDMA	1	C302996	Atari	SDMA CUSTOM CHIPSET		
U37	32MHz	1	MXO45-3C-32M000000	CTD	XTAL OSC XO 32.0000MHZ HCMOS TH	AR14-HZL/01-TT	Assmann
U38	DSP56001	1	C303002	Atari	DIGITAL SIGNAL PROCESSOR		
U39	74ACT157	1	MC74ACT157DR2G	OnSemi	IC MULTIPLEXER4 X2:1 16-SOIC		
U40	74ACT74DR	1	SN74ACT74DR	Texas Instruments	IC FF D-TYPE 2-ELE 1-BIT 14-SOIC		
U41	OSC	1	ECS-100A-251.7	ECS	XTAL OSC XO 25.1750MHZ TTL TH	AR14-HZL/01-TT	Assmann
U42, U43, U59, U60, U61	74LS244	5	MC74HC244ADWR2G	OnSemi	IC BUFF 2V/6V 20-SOIC		
U44	G20V8	1	C014386	Atari	EE PLD 12NS	GAL20V8AS-12HB1	ST Microelectronics
U45, U48, U53	IS65C256AL-25ULA3	3	IS65C256AL-25ULA3	ISSI	IC SRAM 256KBIT PARALLEL 28SOP		
U49, U50	NE556	2	NE556DR	Texas Instruments	IC OSC TIMER DUAL 100KHZ 14-SOIC		
U51	27C4096	1	AT27C4096-55JU	Microchip	IC EPROM 4MBIT PARALLEL 44PLCC	8444-21B1-RK-TP	3M
U52	MC6850	1	EF68B50P	STM	ASYN COMMUNICATION INTERFACE	A24-LC-TT	Assmann
U55	MC68LC030FG	1	SC414230FE16	Motorola	MICROPROCESSOR 32-BIT	MC68LC030FG	Motorola
U56	COMBEL	1	C302995	Atari	COMBEL CUSTOM CHIPSET		
U57	74LS74	1	SN74LS74ADR	Texas Instruments	IC FF D-TYPE 2-ELE 1-BIT 14-SOIC		

Designator	Name	Quantity	Manufacturer Part Number	Manufacturer	Description	Alt Part or Accessory	Alt Manufacturer
U58	74HC04	1	M74HC04YRM13TR	STM	IC INVERTER6CH 1-INP 14SO		
U62	G16V8	1	C302831-001	Atari	EE PLD 7NS	ATF16V8C-7PU	Microchip
U63	G16V8	1	C302832-001	Atari	EE PLD 7NS	ATF16V8C-7PU	Microchip
U64	DS1287	1	DS1287	Dallas	REAL TIME CLOCK NVRAM BATT	NWX287	Benchmark
U65	OSC	1	32.084988 MHz	CT	OSCILLATOR 32.084988	works with 32.00MHz	
U66	74LS373	1	SN74HC373DWR	Texas Instruments	IC D-TYPE TRANSP 8:8 20-SOIC		
U67	G16V8	1	C302830-001	Atari	EE PLD 7NS	ATF16V8C-7PU	Microchip
U68	G16V8	1	C302833-001	Atari	EE PLD 7NS	ATF16V8C-7PU	Microchip
U701	CDC341	1	CDC341DWR	Texas Instruments	IC CLK BUFFER 1:8 80MHZ 20SOIC		
Y1	2.4576MHz	1	ECS-24-32-1X	ECS	CRYSTAL 2.4576MHZ 32PF TH		
Y2	4.43MHz	1	ECS-44-20-1X	ECS	CRYSTAL 4.433618MHZ 20PF TH		
Y3	32M08	1	CO70176	Atari	RESONATOR 32.08MHZ		